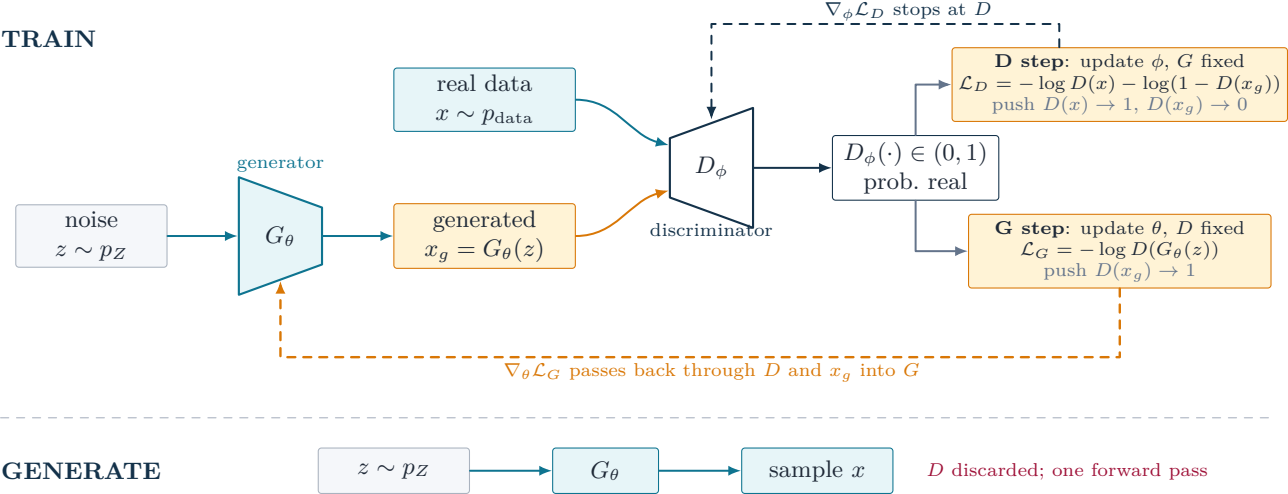
TRAIN
$\nabla_\phi \mathcal{L}_D$ stops at D
real data
$x \sim p_{\text{data}}$
D_ϕ
discriminator
$D_\phi(\cdot) \in (0,1)$
prob. real
D step: update ϕ, G fixed
$\mathcal{L}_D = -\log D(x) - \log(1 - D(x_g))$
push $D(x) \to 1$, $D(x_g) \to 0$
generator
noise
$z \sim p_Z$
G_θ
generated
$x_g = G_\theta(z)$
G step: update θ, D fixed
$\mathcal{L}_G = -\log D(G_\theta(z))$
push $D(x_g) \to 1$
$\nabla_\theta \mathcal{L}_G$ passes back through D and x_g into G
GENERATE
$z \sim p_Z$
G_θ
sample x
D discarded; one forward pass